Founder Memo: COD Risk Decision Engine

COD Risk Decision Dashboard | Reporting period: 1 April – 30 September 2026

To: Founder and Leadership Team **Subject:** Reducing COD RTO risk without unnecessarily losing genuine customers

1. Executive Summary

Cash-on-Delivery (COD) orders create a significant operational and financial challenge. Customers can refuse delivery or become unavailable, resulting in forward shipping, return shipping, packaging and potential product-damage costs. The historical dataset contains 1,988 decision records. After removing duplicate records and excluding orders without final outcomes, 1,987 orders were available for historical evaluation. Of these, 565 resulted in RTO, producing a historical RTO rate of **28.43%**. Using an estimated cost of ■169.40 per RTO, the historical RTO cost is approximately ■**95,711**.

The configurable COD Risk Decision Engine assigns each order one of four recommendations: **ship as-is**, **nudge to prepaid**, **request WhatsApp confirmation**, or **hold for manual review**. Its objective is to reduce avoidable RTO exposure while protecting genuine customers, particularly first-time customers acquired through paid marketing channels.

2. What the Decision Engine Does

The engine evaluates previous order and RTO history, customer type (first-time or returning), phone-number validity, pincode validity, city availability and marketing source. It applies configurable rules rather than relying on an LLM to calculate risk or make financial decisions. The current configuration uses a prior-RTO threshold of one and identifies Meta Ads first-time customers as eligible for a prepaid nudge. Missing or invalid information generally triggers WhatsApp confirmation, while inconsistent customer-history records are sent for manual review.

Initial decision	Orders
Ship as-is	1,251
Nudge to prepaid	550
WhatsApp confirmation	183
Hold for manual review	4
Total	**1,988**

Hold for manual review is not automatic cancellation. Human review is required before any cancellation decision.

3. Historical Backtesting Findings

A chronological 80:20 split was used to examine the decision groups across an earlier training period and a later test period.

Decision	Training RTO	Test RTO
Ship as-is	24.82%	21.15%
Nudge to prepaid	32.59%	39.60%
WhatsApp confirmation	44.52%	24.32%
Hold for manual review	100% (4 orders)	No test records

The test period contained 398 orders, of which 104 resulted in RTO, giving an overall RTO rate of 26.13%. This was 2.88 percentage points lower than the training-period rate of 29.01%. These results show different historical RTO patterns across decision groups, but do not prove that the engine caused an improvement. The rules were developed after examining the full dataset, so this is not independent prospective validation. The manual-review group also contains only four orders, too few for a reliable performance conclusion.

4. Financial Opportunity: Prepaid Conversion

A hypothetical prepaid-conversion scenario estimates the potential financial opportunity. At the dashboard's default assumptions, the eligible group contains 548 COD orders, with a historical COD RTO rate of 33.94%. The scenario assumes a 20% prepaid conversion rate, a 4% prepaid RTO rate and a ■20 discount per converted order.

■22,294.34

Estimated net financial impact under the selected assumptions

Component	Estimate
RTO cost avoided	■5,559.03
Additional gross margin	■18,927.31
Discount expense	–■2,192.00
Net estimated impact	**■22,294.34**

This is a hypothetical scenario, not measured savings. It assumes converted orders experience the assumed prepaid RTO rate and that RTOs prevented become delivered orders. Payment gateway fees, WhatsApp costs and other operational expenses are not included.

5. Balancing Operations and Marketing

Operations: Use confirmation and manual review to investigate orders with missing, invalid or inconsistent information.
Marketing: Avoid automatically rejecting first-time customers simply because they came through paid acquisition channels. Offer prepaid payment as an option instead.
Customer experience: Allow customers to confirm or decline COD orders through a simulated WhatsApp workflow.

The current mock workflow supports YES, NO, invalid responses and a 24-hour timeout. These interactions are simulated; no actual customer messages have been sent.

6. Risks and Monitoring

- **Data quality:** Invalid phone numbers, missing pincodes and inconsistent customer-history records can affect decisions.
- **False positives:** Genuine customers may be unnecessarily asked to confirm or pay in advance.
- **False negatives:** Risky orders may still be shipped.
- **Small samples:** Some customer and location groups have insufficient records for reliable conclusions.
- **Financial assumptions:** Prepaid conversion, RTO rates and discounts must be validated using real operational results.
- **Privacy:** Customer contact and address information must be protected and excluded from publicly shared logs.

The dashboard currently records zero actual cancellation actions. Therefore, lost gross margin from genuine cancelled orders cannot yet be measured. A future version should record the human review outcome and actual cancellation status.

7. Proposed Next Two Weeks

Period	Priority
Week 1	Validate decision rules, review false positives and false negatives, improve customer-response handling, and add human review outcomes.
Week 2	Run a controlled pilot, monitor confirmation rates and actual prepaid payments, compare RTO outcomes, and update financial assumptions.

The pilot should measure actual RTO rate, prepaid conversion, customer drop-off, cancellation rate, gross margin impact and operational costs before wider deployment.

Conclusion

The COD Risk Decision Engine provides a configurable and explainable framework for identifying potentially risky orders and selecting appropriate actions. Its current historical analysis identifies opportunities for better COD management, while the prepaid scenario demonstrates a possible financial benefit under explicit assumptions. The next step is not immediate full-scale deployment. It is controlled testing with real customer responses, verified payments and actual delivery outcomes.